

GROWING BONES

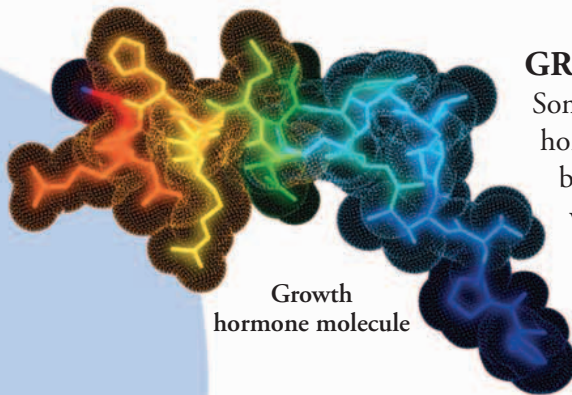
Growth hormone makes you taller by making your bones grow longer. Your bones are too hard to simply stretch in length, but they contain areas of soft tissue near the ends where new bone cells are laid down. Called growth plates, these look like gaps on X-rays. When you reach adulthood and your bones stop growing, they fill up with solid bone and disappear.



► **CHILD'S HAND**
In this 10-year-old's hand, the long bones are still growing at the growth plates, which look like gaps in the X-ray.



► **ADULT'S HAND**
In this 20-year-old's hand, the bones have stopped growing and the growth plates have filled with bone.



GROWTH DRUG

Some children don't produce enough growth hormone and so are shorter than usual. They can be treated with artificial growth hormone (left), which is produced by genetically engineered bacteria. In the past, natural growth hormone taken from the brains of corpses was used, but this treatment carried a risk of disease and was stopped in 1985.

WOW!

Growth hormone makes every organ in the body grow larger except for the brain.

DWARFISM AND GIGANTISM

Problems with the pituitary gland in childhood can have dramatic effects on growth. A lack of growth hormone slows down bone growth, resulting in dwarfism, a condition in which someone is shorter than usual. Too much growth hormone, which can be caused by an overactive pituitary gland, results in gigantism, a condition in which someone grows very tall. This is what happened to Robert Wadlow, an American man who grew to 8 ft 11 in (2.72 m) by age 22, making him the tallest person ever.

► **ROBERT WADLOW**
At age 13, Robert Wadlow was 7 ft 4 in (2.2 m) tall, making him far taller than his father and nearly twice the height of his 9-year-old brother.

